

Information Security

Unit- 7 Network Security

Network Security

- Network security is the practice of protecting a computer network from unauthorized access, misuse, or attacks. It involves using tools, technologies, and policies to ensure that data traveling over the network is safe and secure, keeping sensitive information away from hackers and other threats.

How Network Security Works :

- Network security uses multiple layers of protection both at the network's edge and within it. Each layer has rules and controls to manage who can access network resources. Authorized users can use the network safely, while unauthorized users are blocked.

Levels of Network Security:

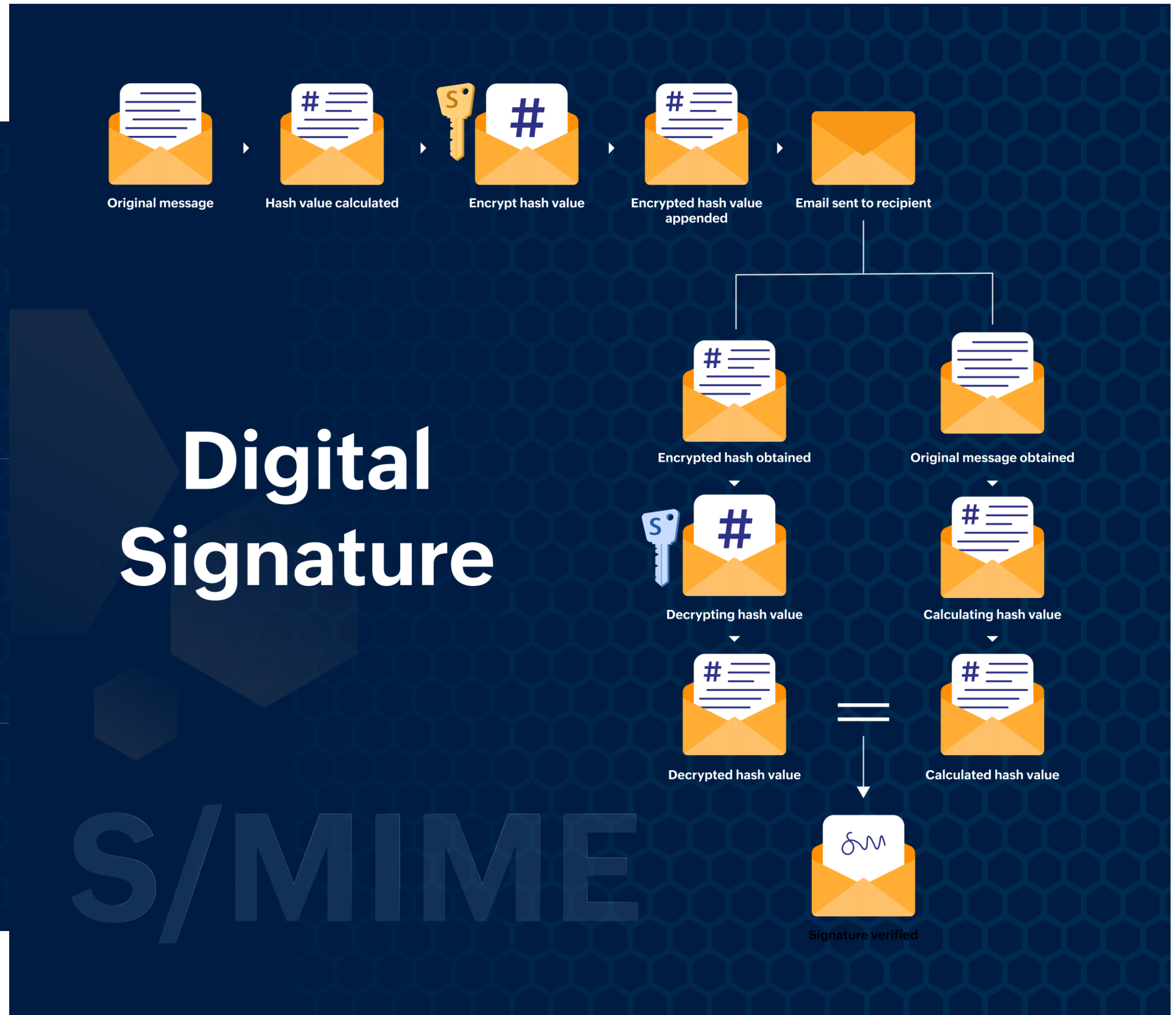
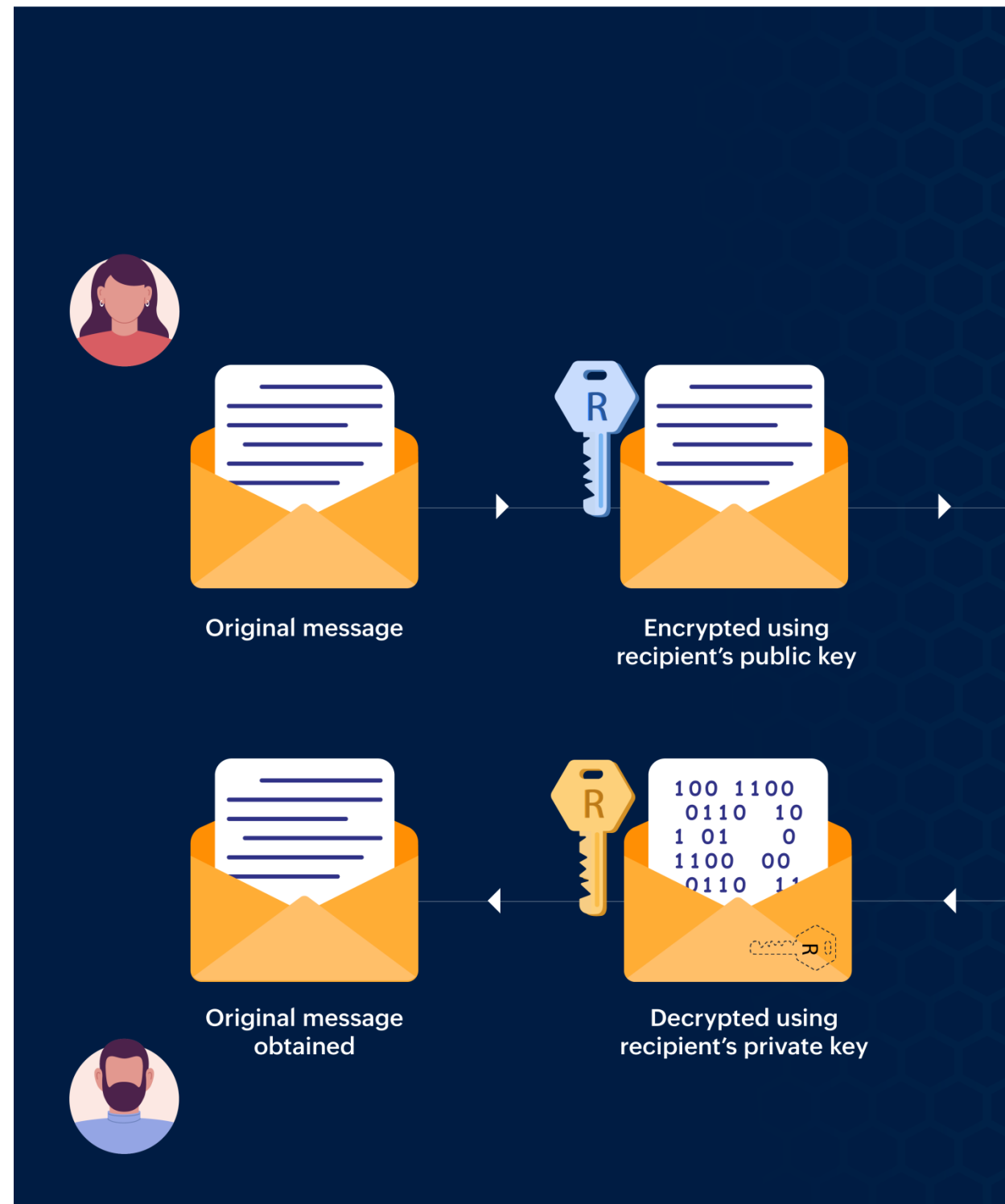
- **Physical Network Security:** Protects the network's physical infrastructure from unauthorized access using devices like biometric systems.
- **Technical Network Security:** Safeguards data stored in or traveling through the network from unauthorized users and malicious activities.
- **Administrative Network Security:** Controls user behavior and permissions, ensuring proper authorization processes and making necessary adjustments to the network's infrastructure to prevent attacks.

Email Security:

- Basically, Email security refers to the steps where we protect the email messages and the information that they contain from unauthorized access, and damage.
- It involves ensuring the confidentiality, integrity, and availability of email messages, as well as safeguarding against phishing attacks, spam, viruses, and another form of malware.
- It can be achieved through a combination of technical and non-technical measures including:
 1. Strong Passwords
 2. Two-factor Authentication
 3. Password rotations
 4. Spam filters
 5. Anti-virus/ Anti-spam applications
 6. Intrusion detection system etc.

S/MIME

- S/MIME, or Secure Multipurpose Internet Mail Extension, is an email encryption and signing industry standard widely used by corporations to enhance email security. S/MIME is compatible with most enterprise email clients.
- In simple terms, S/MIME is an encryption protocol used to digitally sign and encrypt an email to ensure that the email is authenticated and its content is not altered.
- S/MIME works based on asymmetric encryption. This means that this protocol uses a two-key system (Public and Private) that is mathematically related but different, to encrypt and decrypt an email.
- A S/MIME certificate needs to be installed on the email clients of both the recipient and the sender to ensure email encryption at both ends. When an email is sent, the sender encrypts the email using the recipient's public key and the recipient decrypts the email using the private key.
- S/MIME also attaches a digital signature to an email. This ensures that the sender is authorized to send emails from a specific domain.



Advantages of S/MIME

1. It offers verification.
2. It offers integrity to the message.
3. By the use of digital signatures, it facilitates non-repudiation of origin.
4. It offers seclusion.
5. Data security is ensured by the utilization of encryption.
6. Transfer of data files like images, audio, videos, documents, etc. in a secure manner.

Disadvantages of S/MIME

1. Not all email software supports S/MIME signatures.
2. Due to the requirement of implemented certificates all users cannot take benefits of S/MIME as some users only want encryption.

Services of S/MIME

1. Digital Signature, which can maintain data integrity.
2. S/MIME can be used in encrypting messages.
3. By using this we can transfer our data using an e-mail without any problem.

Pretty Good Privacy (PGP)

- PGP is an open-source, freely available software package for email security.
- PGP is a data encryption and decryption program that provides cryptographic privacy and authentication for data communication.
- It is often used for signing, encrypting, and decrypting texts, emails, files, directories, and whole disk partitions and to increase the security of e-mail communications.
- Phill Zimmermann developed PGP in 1991 AD.

Service Provided by PGP

- The actual operation of PGP consists of 5 services:

1. Authentication :

- PGP provides authentication through the use of digital signature.
- A hash code of message is created using SHA-1. The hash code is encrypted using DSS or RSA with the sender's private key including message. The receiver uses sender's public key to decrypt and recover the hash code. The receiver generates a new hash code for the message and compares it with the decrypted hash code. If the two match, the message is accepted as authentic.

2. Confidentiality :

- PGP provides confidentiality through the use of symmetric block encryption. A message is encrypted using CAST- 128 or IDEA or 3DES with one-time session key generated by the sender. The session key is encrypted using with the recipient's public key and included with the message. The receiver uses RSA with its private key to decrypt and recover the session key. The session key is used to decrypt the message.

3. Compression :

- PGP compresses the message after applying the signature but before encryption. This has the benefits of saving space both for email transmission and for file storage. Message encryption is applied after compression to strengthen cryptographic security. To compress the data it uses ZIP algorithm.

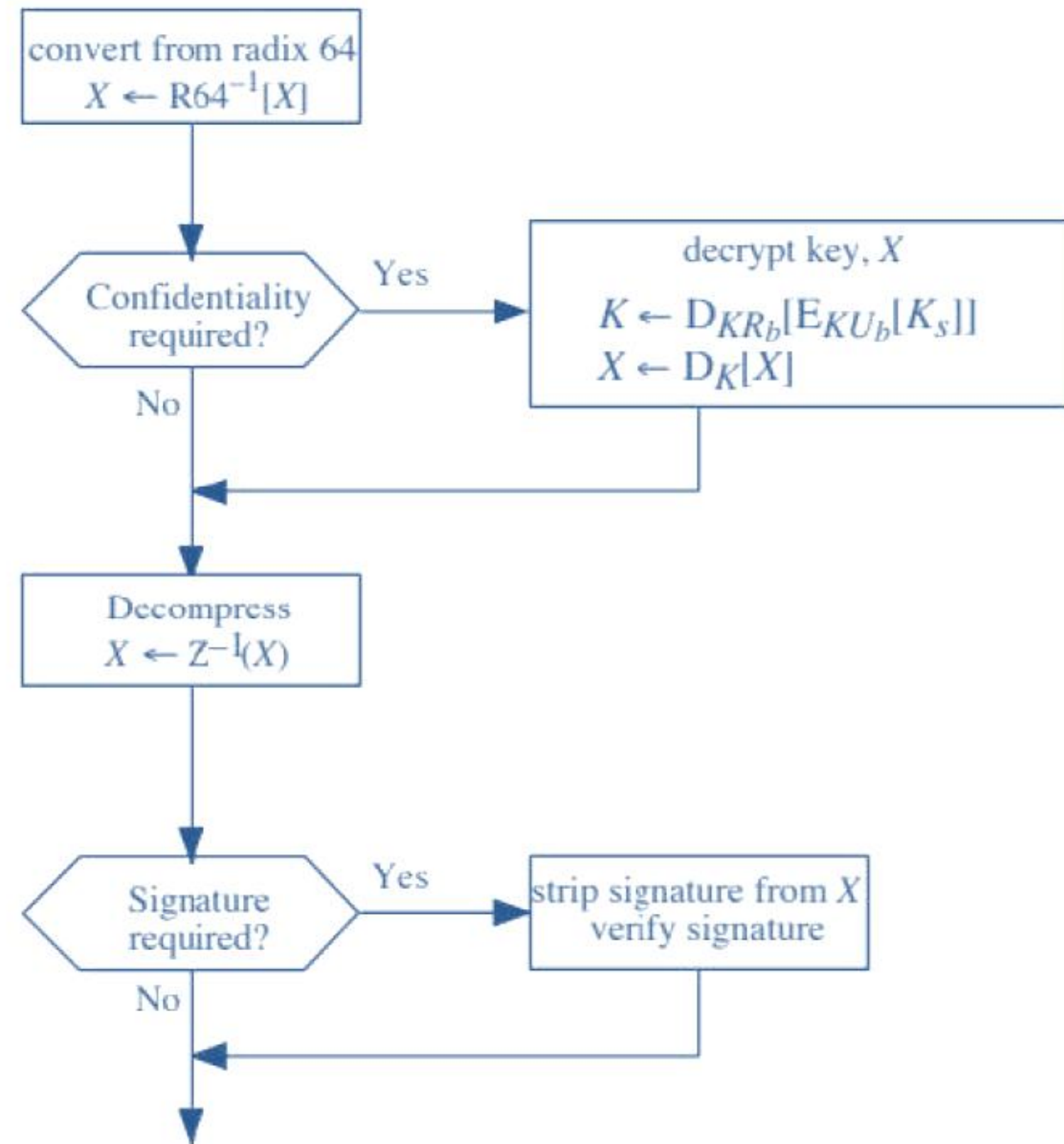
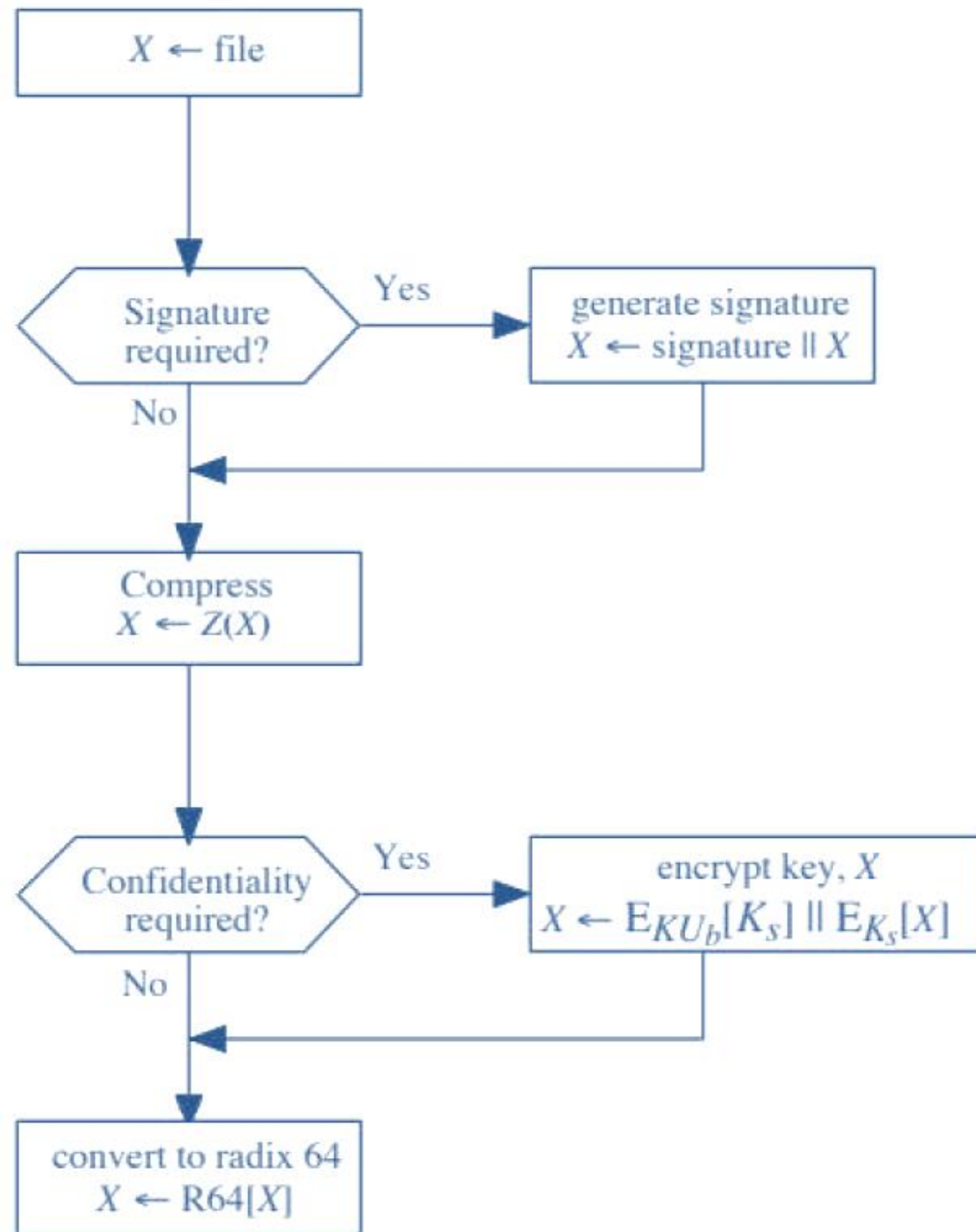
4. Email Compatibility :

- The scheme used for e-mail compatibility is radix-64 conversion. To provide transparency for email application, an encrypted message may be converted to an ASCII string using radix 64-conversion. The use of radix 64 expands a message by 33%.

5. Segmentation:

- Email facilities often are restricted to a maximum length. To accommodate this, PGP automatically subdivides a message that is too large into segments that are small enough to send via e-mail. The segmentation is done after all of other processing, including the radix-64 conversion.

Transmission and Reception of PGP Message



Transmission and Reception of PGP Message

- On transmission, if it is required, a signature is generated using a hash code of the uncompressed plaintext. Then the plaintext, plus signature if present, is compressed. Next, if confidentiality is required, the block (compressed plaintext) is encrypted and prepended with the public-key-encrypted symmetric encryption key. Finally, the entire block is converted to radix-64 format.
- On reception, the incoming block is first converted back from radix-64 format to binary. Then, if the message is encrypted, the recipient recovers the session key and decrypts the message. The resulting block is then decompressed. If the message is signed, the recipient recovers the transmitted hash code and compares it to its own calculation of the hash code.

Secure Socket Layer (SSL) Protocol

- It is an internet protocol for secure exchange of information between a web browser and a web server. SSL encrypt the link between a web server and a browser which ensures that all data passed between them remains private and free from attack.
- It is mainly designed to provide security and compression services to data generated form the application.
- Typically, SSL can receive data form the application layer protocol, but the protocol is HTTP.
- The data received form the application is compressed (optional), signed and encrypted.
- The data is then passed to reliable transport layer protocol such as TCP.

Purpose of SSL

- SSL provides confidentiality, authentication and data integrity in internet communication. SSL is the predecessor to the modern TLS encryption used today.



- In above figure user interact with server but server uses only normal http, it shows unsecured connection between server and client. Hacker or attacker can capture the message from unsecured connection. Username and password are also sent in plain text form. It means hacker or attacker get user's sensitive information.



- In above figure user interact with server and server user secure https, it shows secure connection between client and server. SSL certificate is added at server side. So, http use SSL, it converts into https. Connection is secured, it means all the data is transmitted in encrypted form during transmission. Hacker or attacker cannot get any information from that connection.

Service Provided By SSL

- SSL provides several services on data received from the application layer.

1. Fragmentation :

- First, SSL divides the data into block of 2^{14} bytes or less.

2. Compression :

- Each fragment of data is compressed using one of the lossless compression methods. The service is optional.

3. Message Integrity :

- To preserve the integrity of data, SSL uses a keyed hash function to create a MAC.

4. Confidentiality :

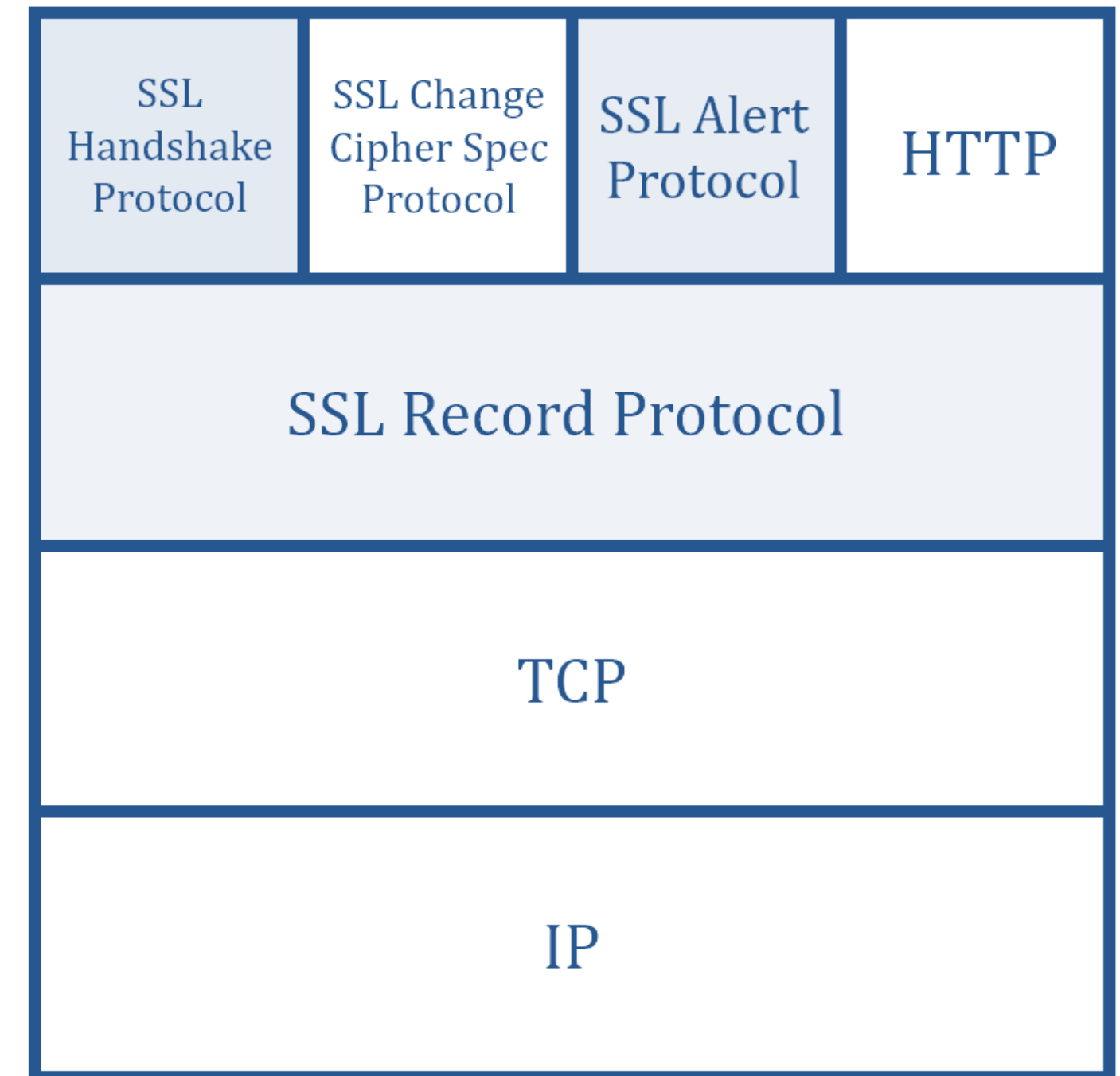
- To provide confidentiality, the original data and MAC are encrypted using symmetric key cryptography.

5. Framing :

- A header is added to the encrypted payload, The payload is then passed to a reliable transport layer protocol.

SSL Architecture

- SSL encrypt the data received from application layer of client machine and add its own header (SSL header) into the encrypted data and send encrypted data to the server side.
 - SSL is not a single protocol to perform security tasks. There are two layers of sub-protocols which supports SSL these are the SSL handshake protocol, SSL change cipher specification, SSL alert protocol and the SSL record protocol shown in architecture.
1. **SSL Handshake Protocol:** Connection establishment.
 2. **SSL Change Cipher Spec Protocol:** Use of required cipher techniques for data encryption.



SSL Architecture

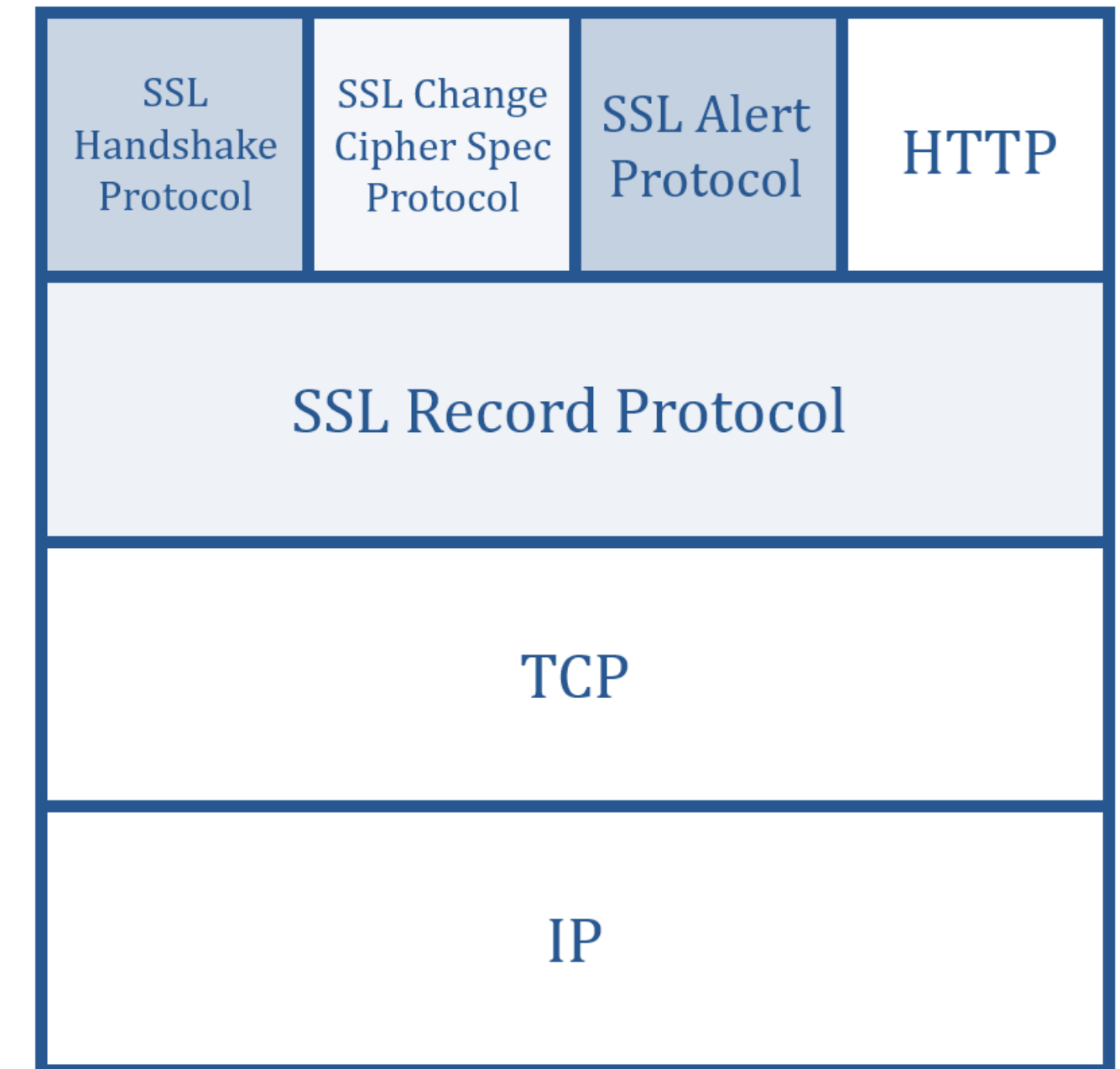
3. **SSL Alert Protocol:** Alert (warning, error if any) generation.

4. **SSL Record Protocol:** Encrypted data transmission and encapsulation of the data sent by the higher layer protocols.

➤ Two other important concept in SSL are –

SSL Connection: It is a transport that provides a suitable type of service. Each connection is associated with one SSL session.

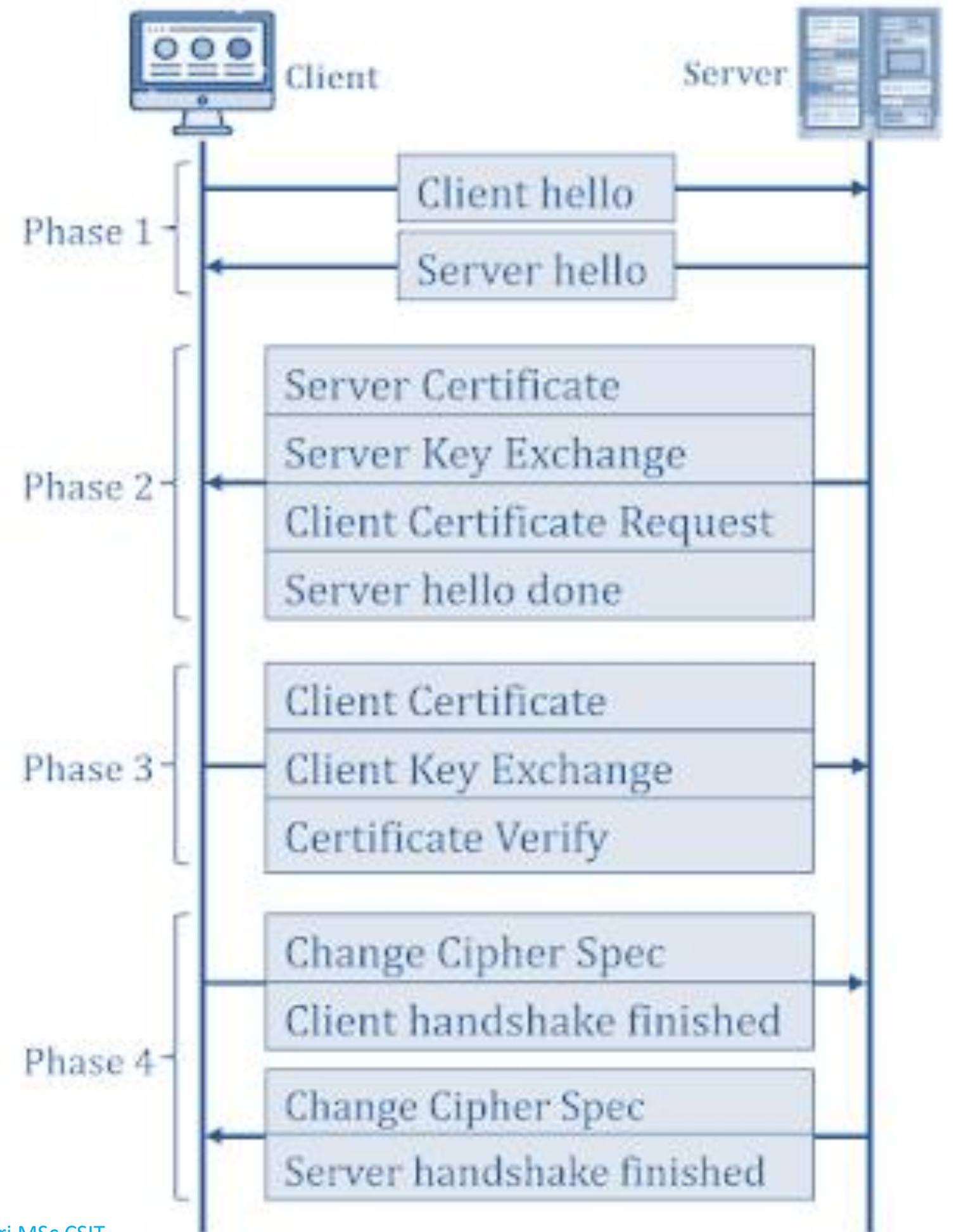
SSL Session: It is a set of cryptographic security parameters which can be shared among multiple SSL connections. An SSL session is an association between a client and a server.



SSL Protocols

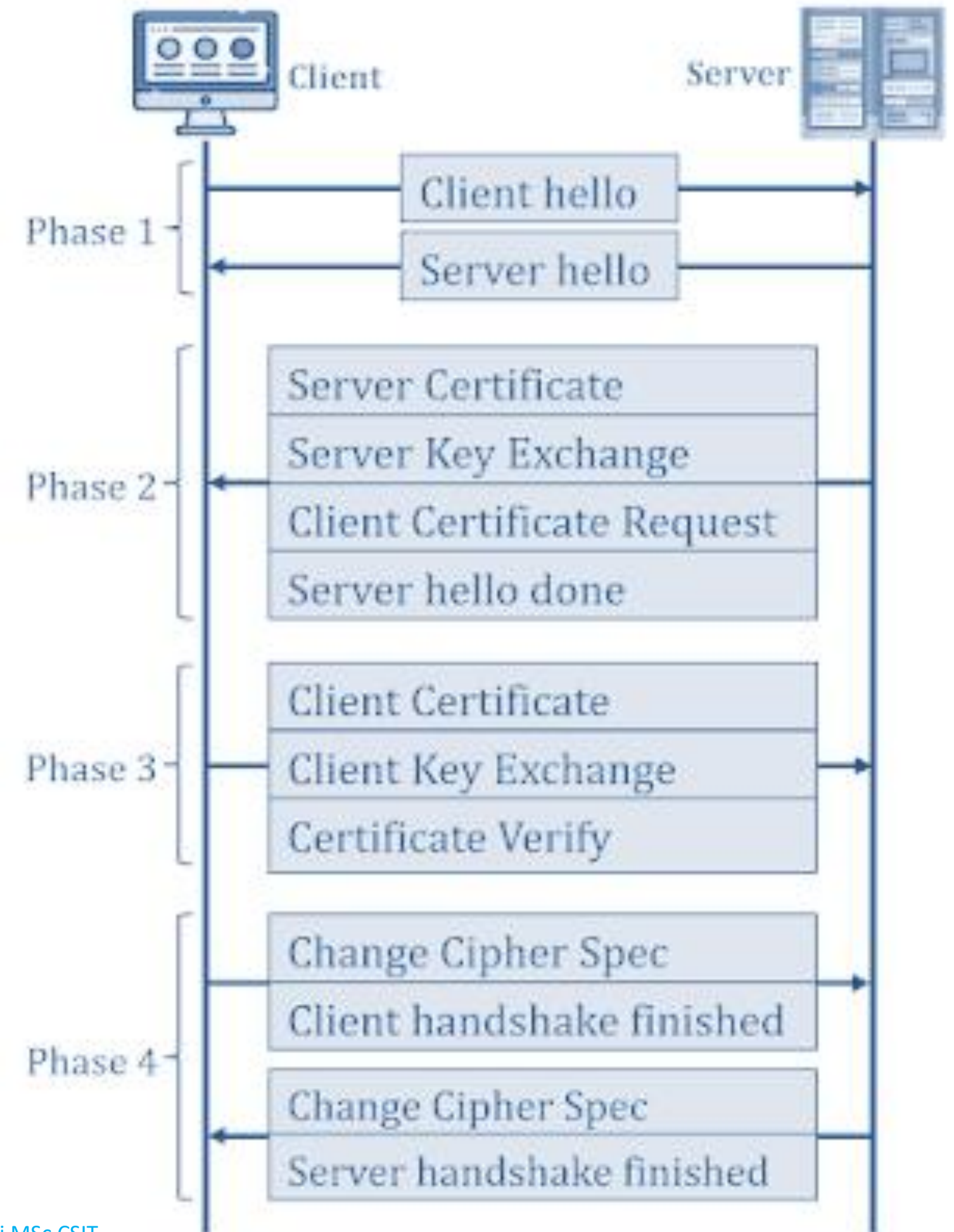
1. SSL Handshake Protocol :

- Handshake protocol is used to establish a secure session between the client and the server. This protocol allows the server and client to authenticate each other and to negotiate an encryption, decryption and MAC algorithm and cryptographic keys to be used to protect data sent in an SSL record. The handshake protocol is used before any application data is transmitted. Handshake protocol uses four phases to complete its cycle.



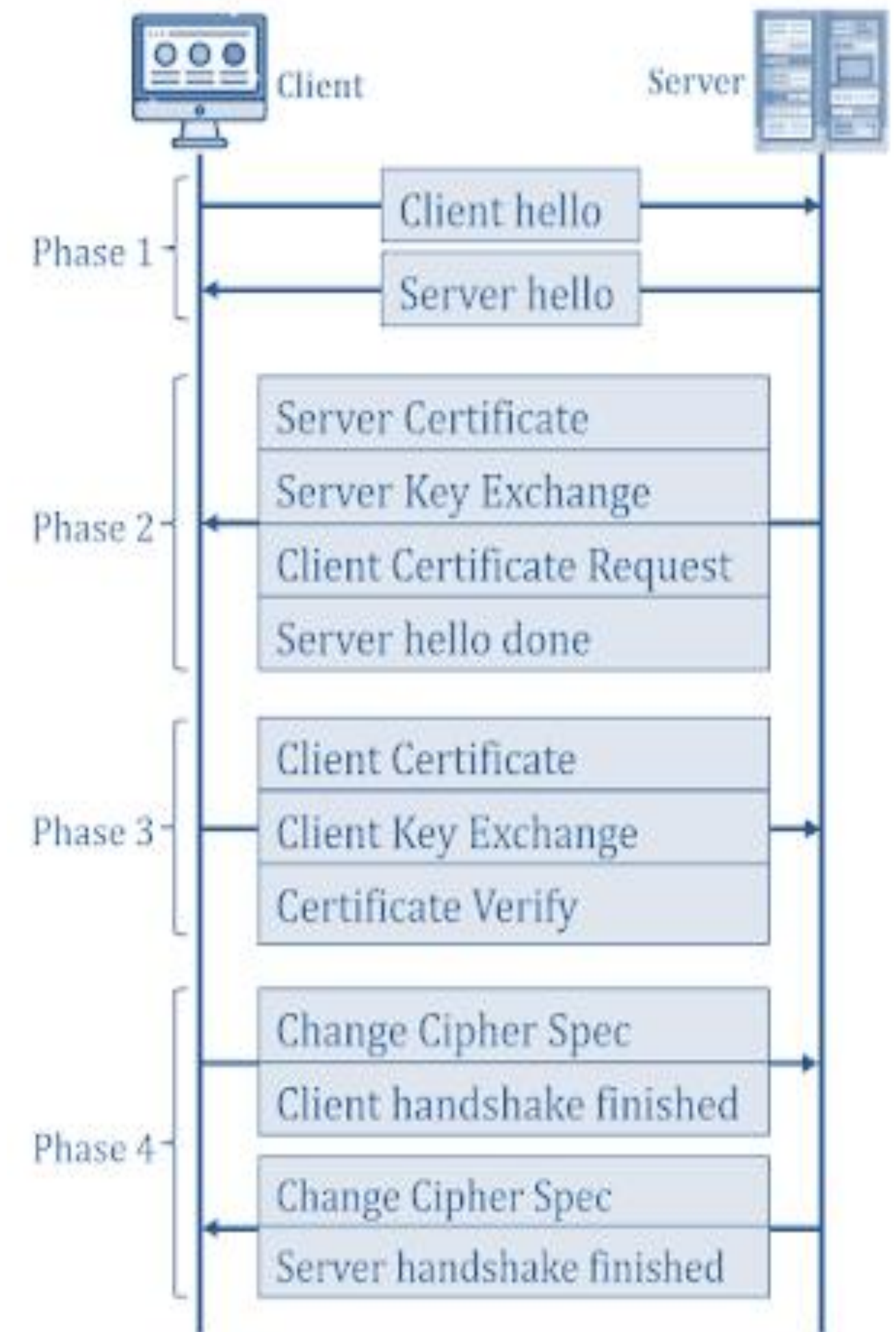
Phase -1 : Establish Security Capabilities:

- This phase is used to initiate a logical connection and to establish the security capabilities that will be associated with it. The exchange is initiated by the client, which sends a “client hello” message. Based on this “client hello” message server return “server hello” message.
- During this phase the information shared between client and server are-
 - ✓ Protocol version
 - ✓ Session Id
 - ✓ Cipher suite
 - ✓ Compression method
 - ✓ Initial random numbers



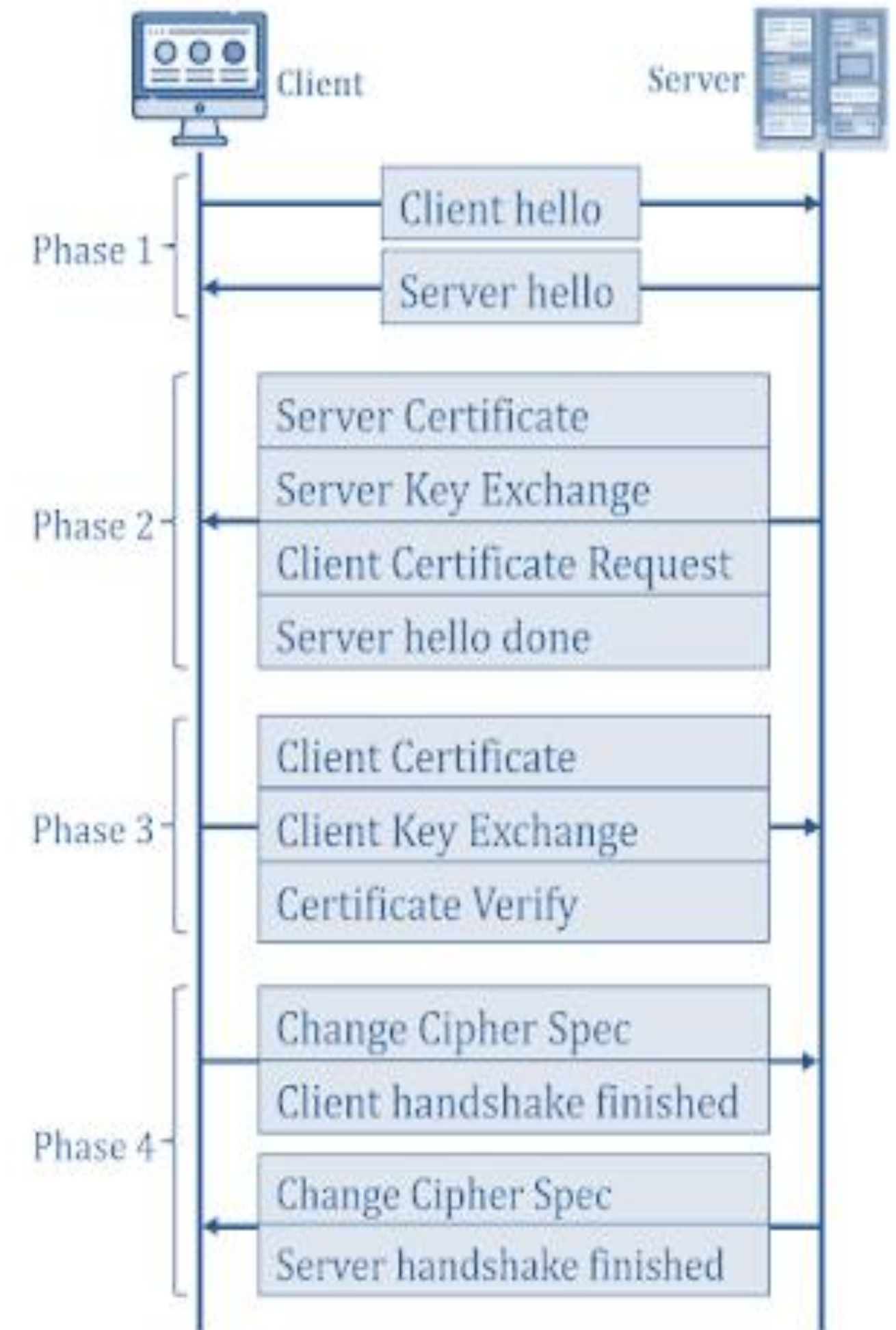
Phase -2 : Server Authentication and key exchange :

- The server sends a certificate message to authentication itself to the client. If the key exchange algorithm is Diffie-Hellman than no need of authentication.
- Server key exchange is optional. It is used only if the server doesn't sends its digital certificate to client.
- The server can request for the digital certificate of client. The client's authentication is optional.
- The server message hello done is the last message in phase 2, this indicates to the client that the client can now verify all the certificates received by the server. After this hello message done, the server waits for the client-side response in phase 3.



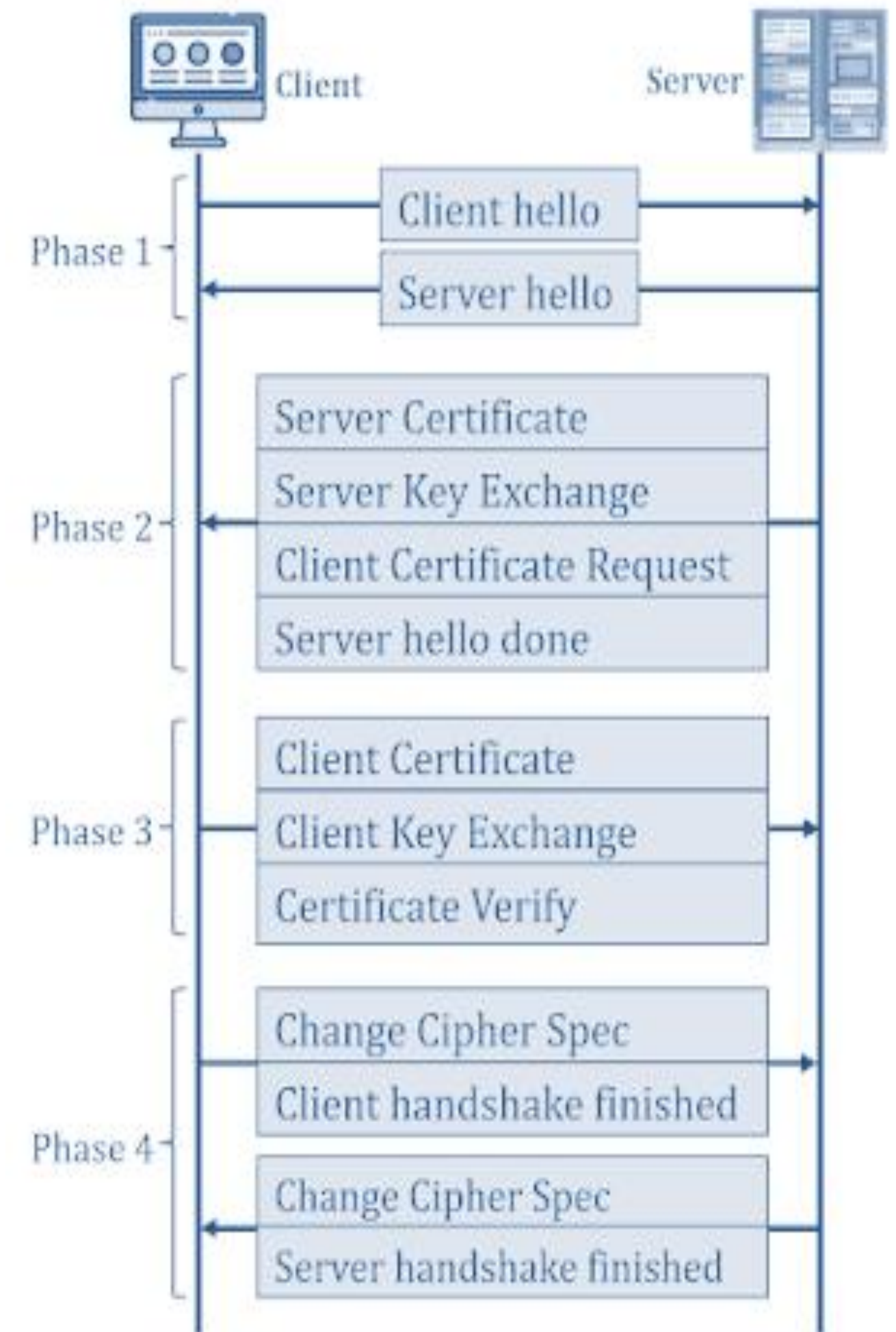
Phase - 3 : Client Authentication and Key Exchange :

- **Client Certificate:** It is optional, it is only required if the server had requested for the client's digital certificate. If client doesn't have certificate, it can be sending no certificate message. Then it is up to server's decision whether to continue with the session or to abort the session.
- **Client key exchange:** The client sends a client key exchange, the contents in this message are based on key exchange algorithms between both the parties.
- **Certificate Verify:** It is necessary only if the server had asked for client authentication. The client has already sent its certificate to the server. But additionally if server wants then the client has to prove that it is authorized holder of the private key. The server can verify the message with its public key already sent to ensure that the certificate belongs to client.



Phase - 4 : Finish

- **Change cipher spec:** It is a client-side messages telling about the current status of cipher protocols and parameters which has been made active from pending state.
- **Finished:** This message announces the finish of the handshaking protocol from client side.
- **Change cipher spec:** This message is sent by server to show that it has made all the pending state of cipher protocols and parameters to active state.
- **Finished:** This message announces the finish of the handshaking protocol from server and finally handshaking is totally completed.



2. SSL Change Spec Protocol :

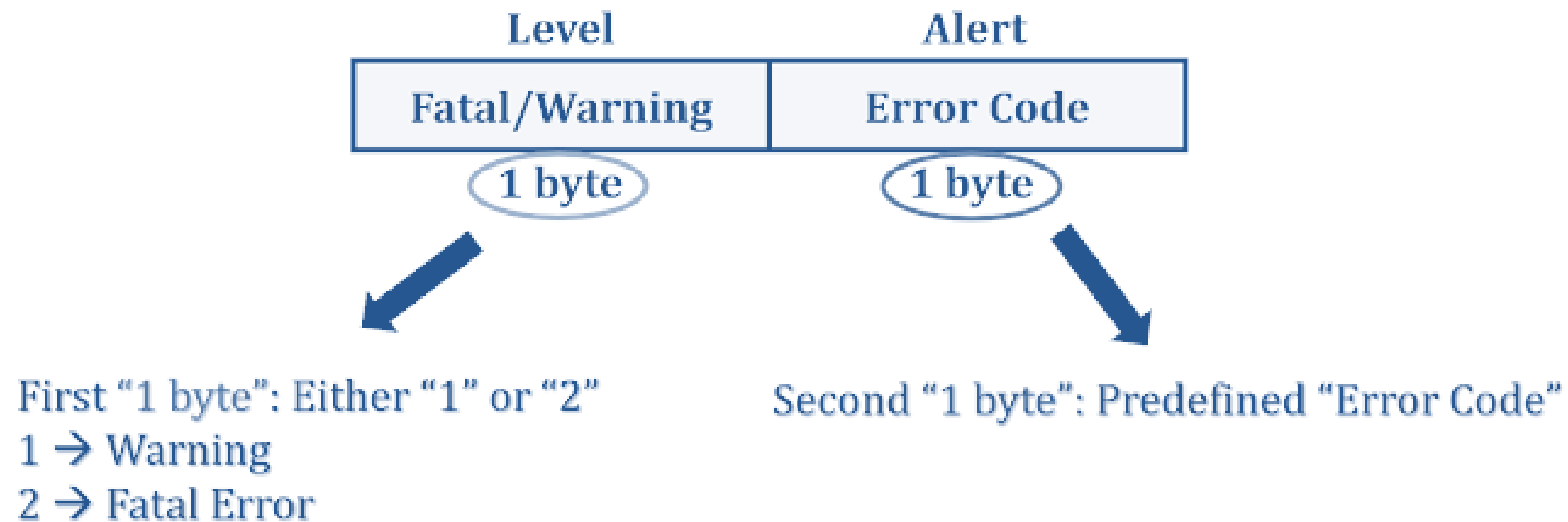
- SSL Change Cipher Spec Protocol is upper layer protocol. It is the simplest protocol. This protocol consists of only single byte with value “1”, as shown in figure. It consists of single message only. It copies pending state to current state, which updates the cipher suite to be used to this connection.

1 byte



3. SSL Alert Protocol :

- SSL uses the Alert protocol for reporting error that is detected by client or server, the party which detects error sends an alert message to other party. If error is serious than both parties terminate the session.
- Figure below shows the Alert Protocol Format



- Alert protocol uses two bytes to generate alert. First 1 byte indicates two values either 1 or 2. "1" value indicate warning and "2" indicate a fatal error.
- Whereas second 1 byte indicates predefined error code either the server or client detects any error it sends an alert containing the error.

3. SSL Alert Protocol :

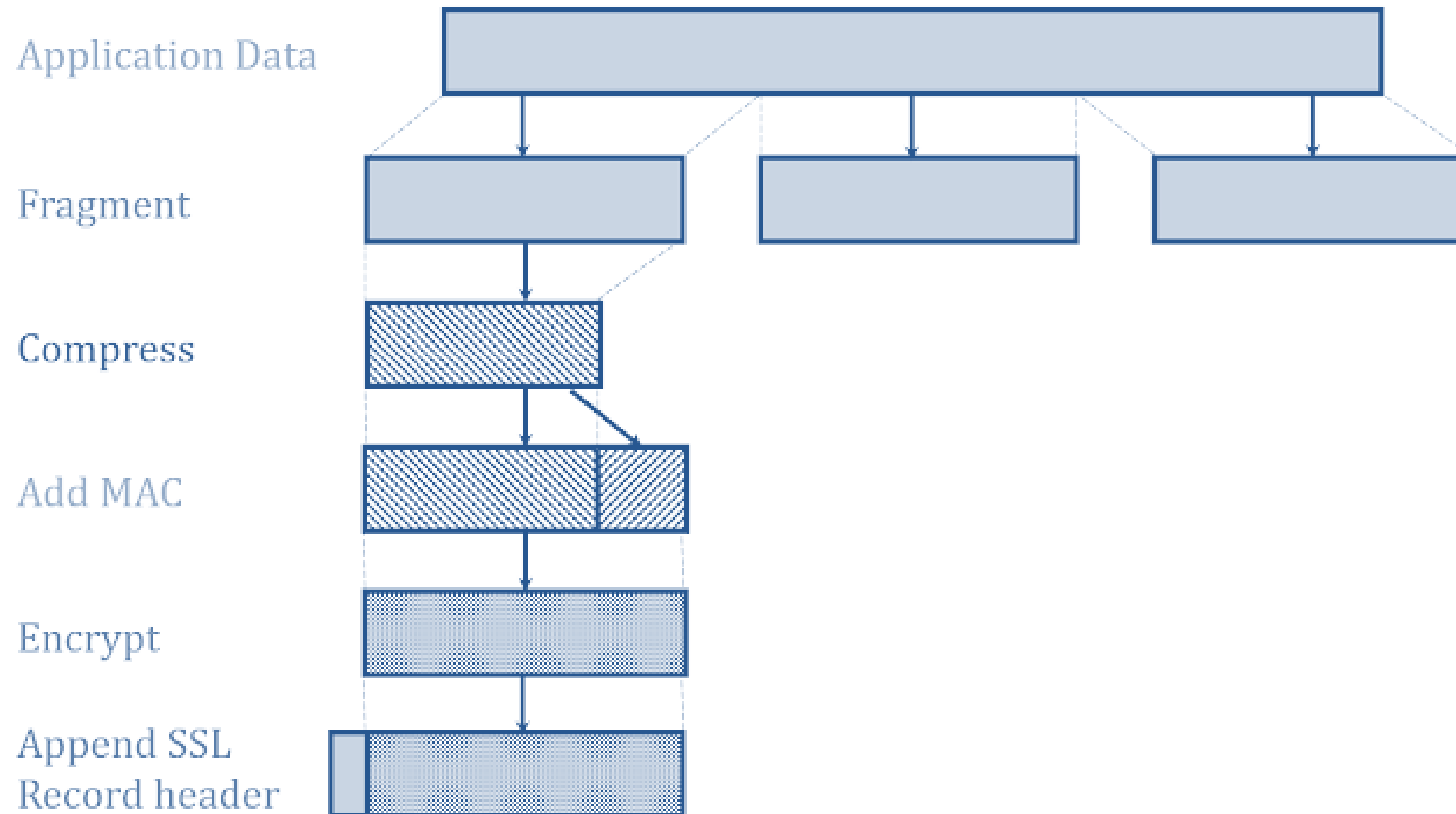
Alert Code	Alert Message	Description
0	close_notify	No more message from sender.
10	unexpected_message	An incorrect message received.
20	bad_record_mac	A wrong MAC received.
30	decompression_failure	Unable to decompress.
40	handshake_failure	Unable to finalize handshake by the sender.
42	bad_certificate	Received a corrupted certificate.
42	No_certificate	Client has no certificate to send to server.
42	certificate_expired	Certificate has expired.

4. SSL Record Protocol :

- After completion of successful SSL handshaking the keen role of SSL record protocol starts now. SSL record protocol is second sub-protocol of SSL also called lower-level protocol. As defined earlier the SSL record protocol is responsible for encrypted data transmission and encapsulation of the data sent by the higher layer protocols also to provide basic security services to higher layer protocols. SSL records protocol provides different service like data authentication; data confidentiality through encryption algorithm and data integrity through message authentication to SSL enabled connections.
- The record protocol provides two services in SSL connection:
 1. **Confidentiality:** This can be achieved by using secret key, which is already defined by handshake protocol.
 2. **Integrity:** The handshake protocol defines a shared secret key that is used to assure the message integrity.

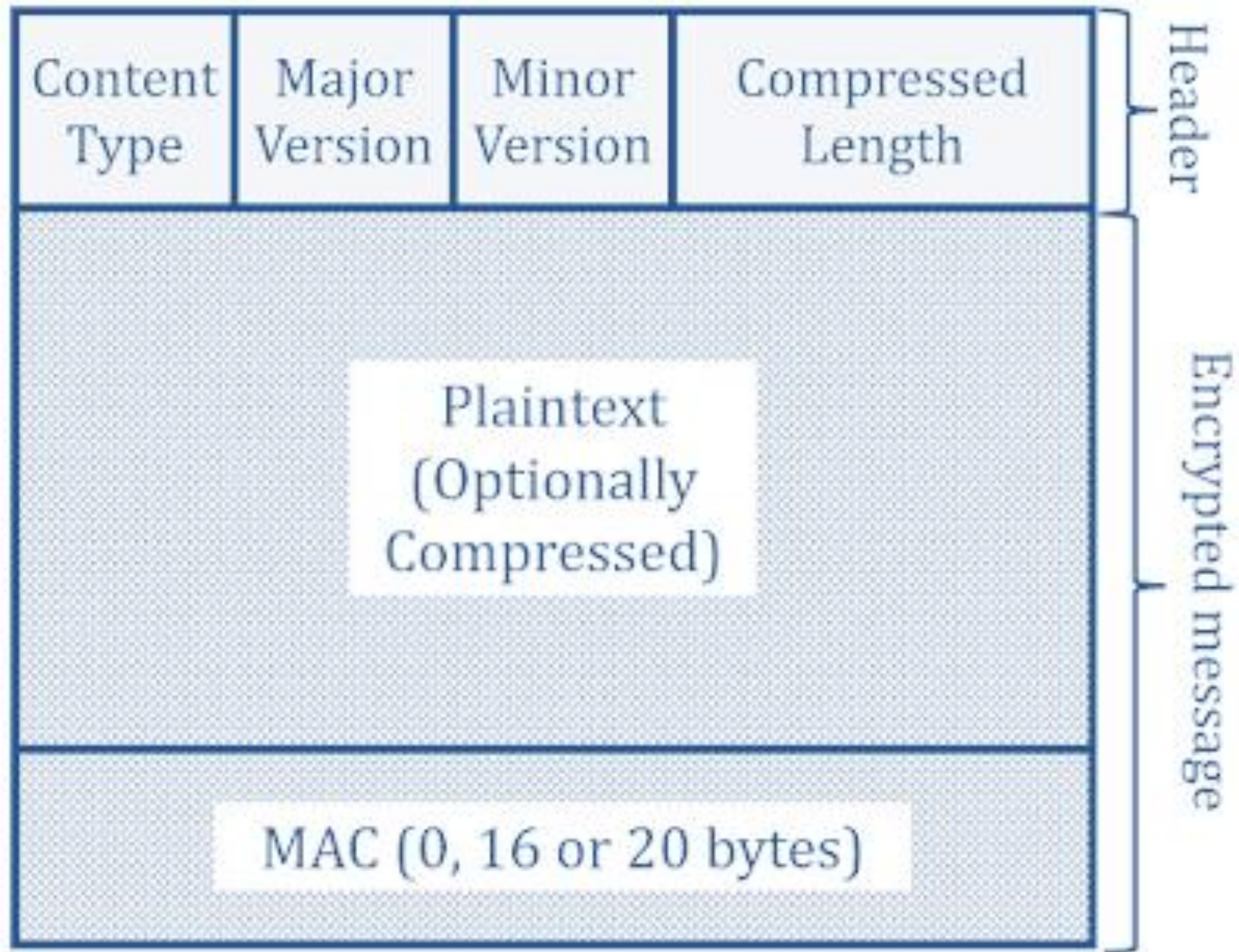
4. SSL Record Protocol :

- Following are the operation performed in Record protocol after connection is established and authentication is done of both client and server.



1. **Fragmentation:** The original message that is to be sent is broken into blocks. The size of each block is less than or equal to 214 bytes.
2. **Compression:** The fragmented blocks are compressed which is optional. It should be noted that the compression process must not result into loss of original data.
3. **Addition of MAC:** A short piece of information used to authenticate a message for integrity and assurance of message.
4. **Encryption:** The overall steps including message is encrypted using symmetric key but the encryption should not increase the overall block size.
5. **Append Header:** After all the above operation, header is added in the encrypted block which contains following fields.
 - a. **Content type:** It specifies which protocol is used for processing.
 - b. **Major version:** It specifies the major version of SSL used, for example if SSL version 3.1 is in use then this field contains 3.
 - c. **Minor version:** It specifies minor version of SSL used, for example version 3.0 is in use then field contains 0.
 - d. **Compressed length:** It specifies the length in bytes of the original plain text block.

Figure shows SSL record protocol header format.



Transport Layer Security (TLS)

- Transport Layer Security, or TLS, is a widely adopted security protocol designed to facilitate privacy and data security for communications over the Internet.
- A primary use case of TLS is encrypting the communication between web applications and servers, such as web browsers loading a website.
- TLS can also be used to encrypt other communications such as email, messaging, and voice over IP (VoIP).
- TLS was proposed by the Internet Engineering Task Force (IETF), an international standards organization, and the first version of the protocol was published in 1999. The most recent version is TLS 1.3, which was published in 2018.

Relation with SSL

- TLS evolved from a previous encryption protocol called Secure Sockets Layer (SSL), which was developed by Netscape.
- TLS version 1.0 actually began development as SSL version 3.1, but the name of the protocol was changed before publication in order to indicate that it was no longer associated with Netscape.
- Because of this history, the terms TLS and SSL are sometimes used interchangeably.

SSL Vs TLS

- **Version:** SSL *3.0* is used where TLS uses *1.0*
- **Cipher suite:** SSL supports *FORTEZZA* but TLS does *not support*.
- **Cryptographic Secrets:** SSL uses *Message Digest to generate master secret* and TLS uses *pseudo random function* to generate master secret.
- **Record Protocol:** For integrity SSL uses *MAC* and TLS uses *HMAC (Hash based MAC)*.
- **Alert Protocol:** TLS *delete* some alert messages of SSL and *add some new* Alert Messages.
- **Certificate Verification:** In SSL it is *complex* and in TLS it is *simple*.
- **Security:** SSL is *less secured* as compared to TLS.

Internet Protocol Security (IPSec)

- IPSec is a group of protocols for securing connections between devices. IPSec helps keep data sent over public networks secure.
- It is often used to set up VPNs, and it works by encrypting IP packets, along with authenticating the source where the packets come from.
- Within the term "IPSec," "IP" stands for "Internet Protocol" and "sec" for "secure." The Internet Protocol is the main routing protocol used on the Internet; it designates where data will go using IP addresses. IPsec is secure because it adds encryption and authentication to this process.
- IP-level security encompasses three functional areas:
 - **Authentication:** The authentication mechanism ensures that the received packet was sent by the identified source. It also assures that the packet has not been altered in transit.
 - **Confidentiality:** The confidentiality facility enables communicating nodes to encrypt messages to prevent eavesdropping by third parties.
 - **Key Management:** It is concerned with secure exchange of keys.

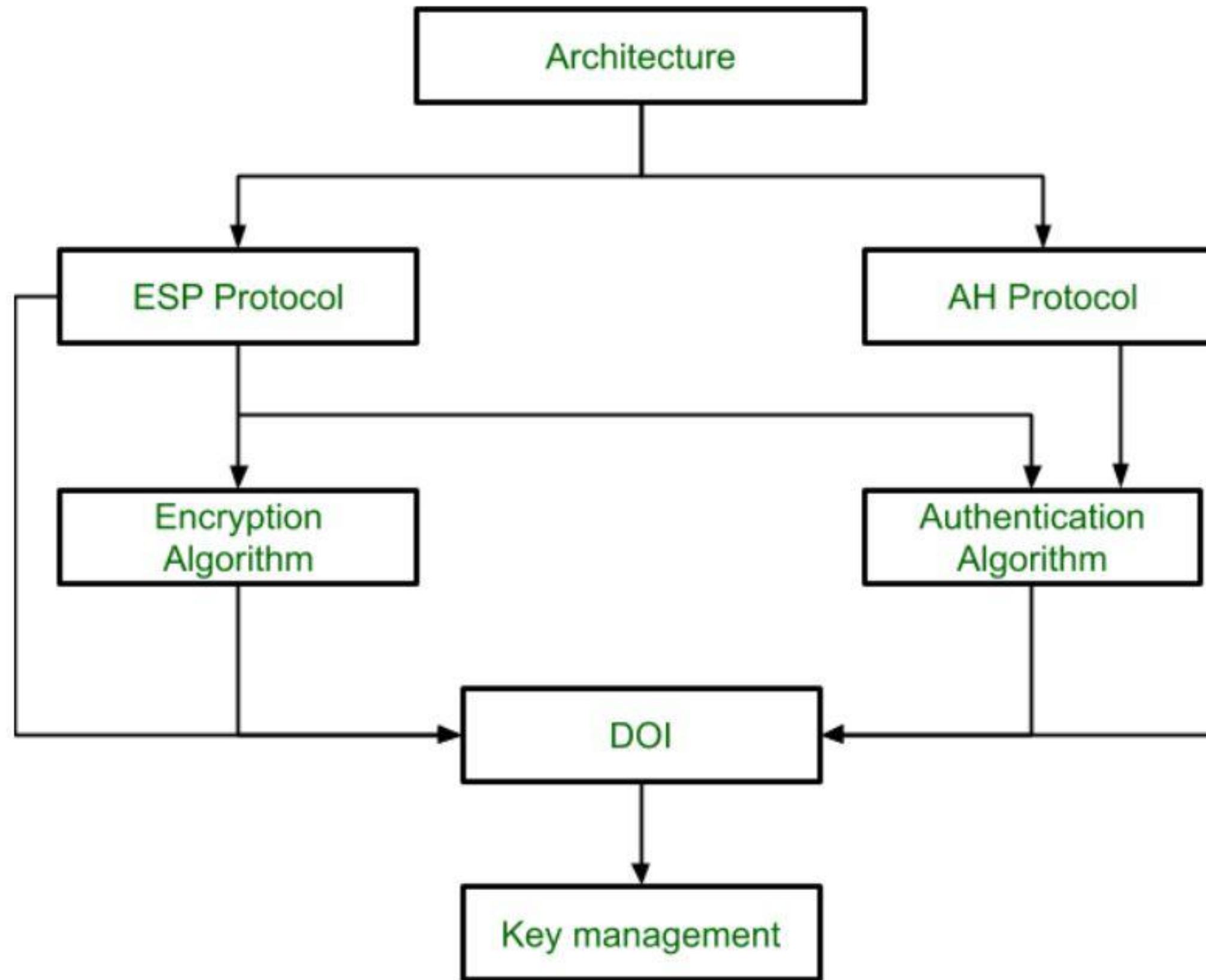
Applications of IPSec

- To secure LAN, private and public WAN, and Internet.
- Secure branch office connectivity over the internet.
- Secure remote access over the Internet.
- Establishing extranet and intranet connectivity.
- Enhancing electronic commerce security.

Benefits of IPSec

- When IPSec is implemented in a firewall or router, it provides strong security to all traffic crossing the perimeter.
- IPSec in a firewall is resistant to bypass if all traffic from the outside must use IP and the firewall is the only means of entrance from the internet into the organization.
- IPSec is below the transport layer(TCP, UDP) and so is transparent to applications.
- There is no need to change software on a user or server system when IPSec is implemented in the firewall or router. Even if IPSec is implemented in end systems, upper-layer software, including applications, is not affected.
- IPSec can be transparent to end users.
- IPSec can provide security for individual users if needed.

IPSec Architecture



1. Architecture :

- Architecture or IP Security Architecture covers the general concepts, definitions, protocols, algorithms, and security requirements of IP Security technology.

2. ESP Protocol :

- (Encapsulating Security payload) Covers packet format and general issues related to the use of the ESP for packet encryption and, optionally, authentication.

3. Authentication Header (AH) :

- Covers packet format and general issues related to the use of the AH for packet authentication.

4. Encryption Algorithm :

- A set of documents that describe how various encryption algorithm are used for ESP

5. Authentication Algorithm :

- A set of documents that describe how various authentication algorithms are used for AH and for authentication option of ESP.

6. Key Management :

- Document that describe key management schemes

7. Domain of Interpretation (DOI):

- Contains the values needed for the other documents to relate to each other.

IPSec Security Services

1. Connectionless Integrity :

- Assurance that received traffic has not been modified. Integrity includes anti-reply defense.

2. Data Origin Authentication:

- Assurance that traffic is sent by legitimate party or parties.

3. Confidentiality :

- Assurance that user's traffic is not examined by non-authorized parties.

4. Access Control :

- Prevention of unauthorized use of a resource.

Operation Mode

- IPSec has two operation mode: transport mode and tunnel mode.

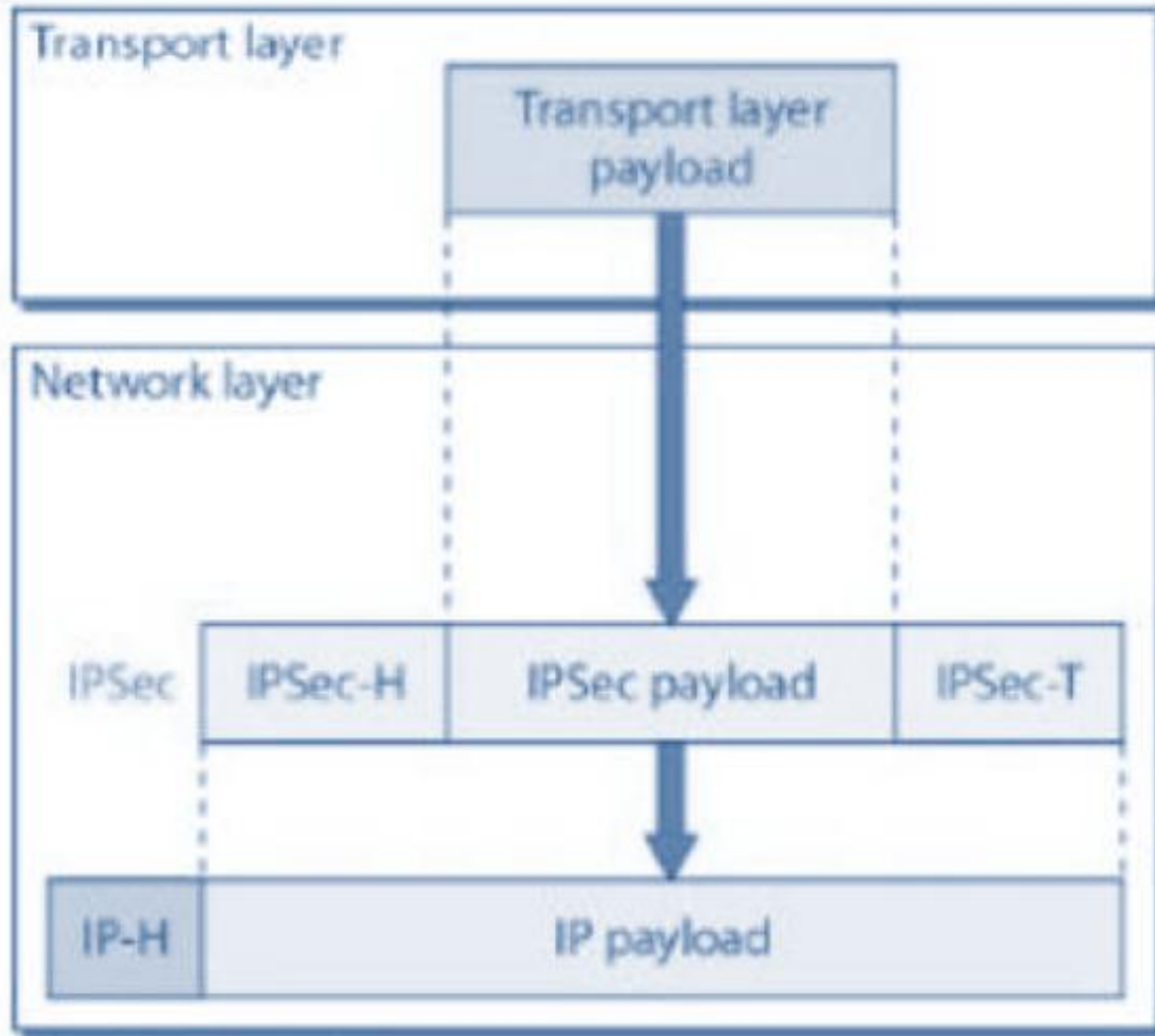
1. Transport Mode :

- In transport mode, only the payload or data of the original IP packet is protected (Encrypted, Authenticated, or both). The protected payload is then encapsulated by the IPSec headers and trailers while the original IP header remains intact and is not protected by IPSec.

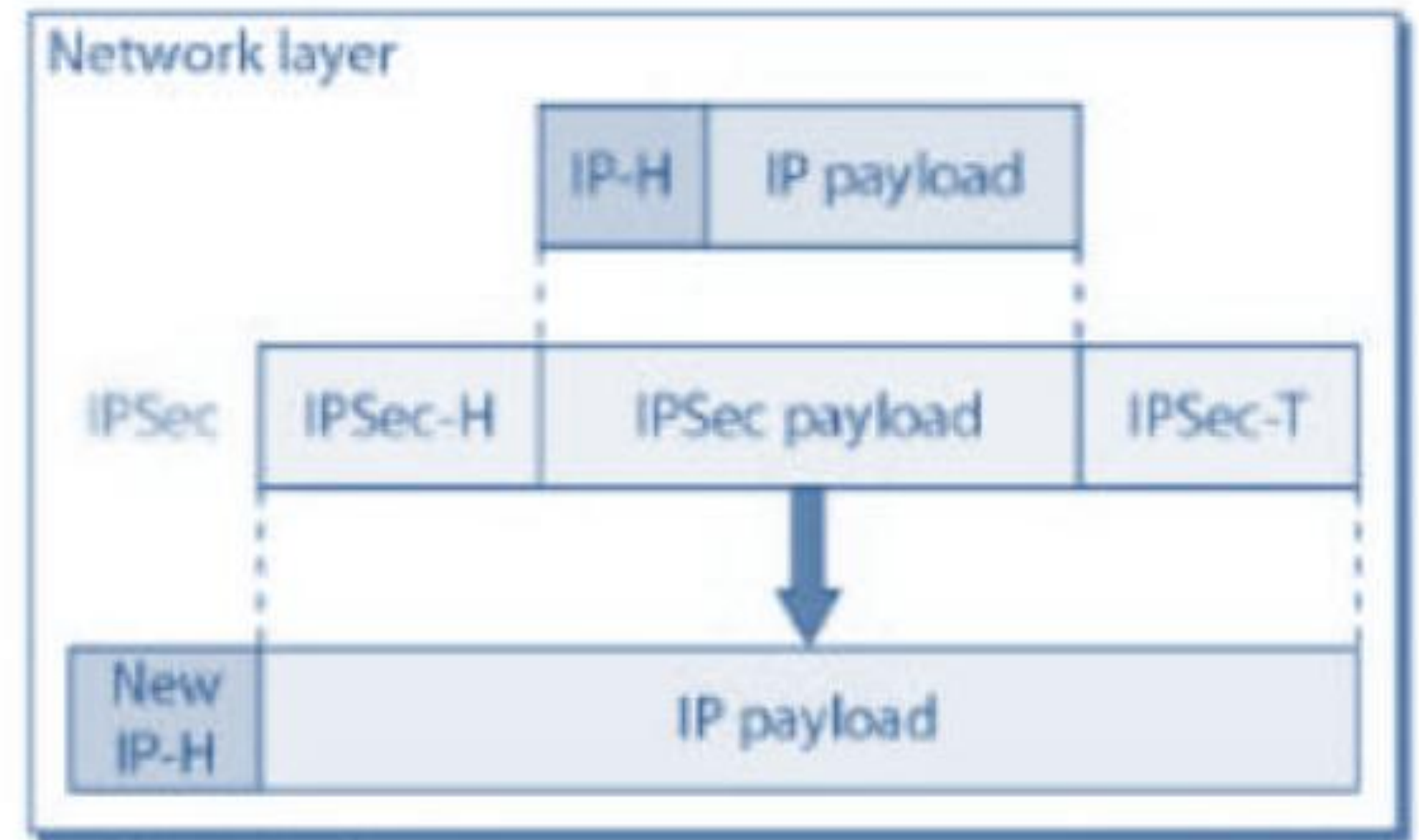
2. Tunnel Mode :

- The entire original IP packet is protected (encrypted, authenticated, or both) in tunnel mode. The packet is then encapsulated by the IPSec headers and trailers. Finally a new IP header is prefixed to the packet, specifying the IPSec endpoints as the source and destination.

Operation Mode



a. Transport mode



b. Tunnel mode

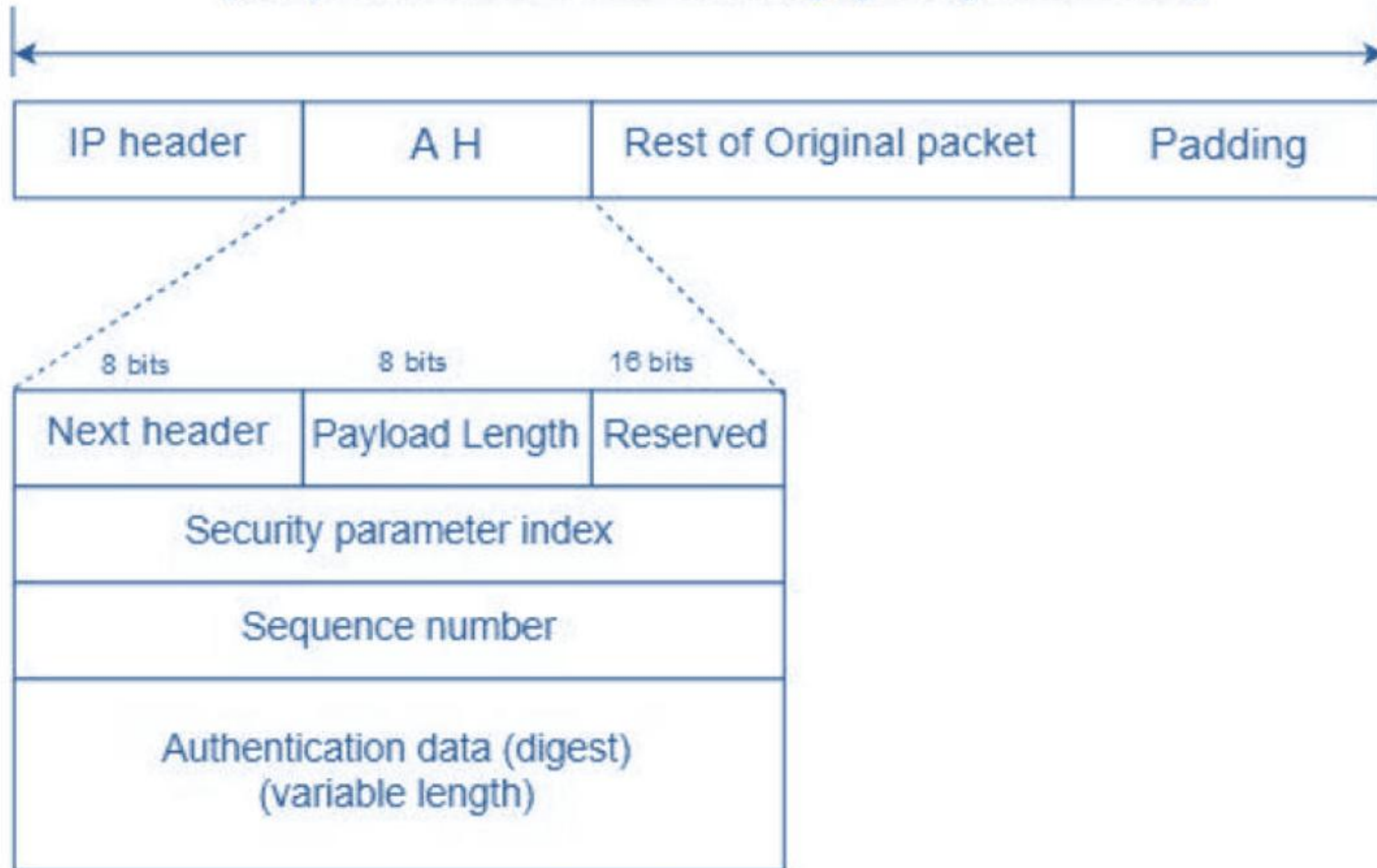
IPSec Protocols (**Self Study**)

- IPSec uses two distinct protocols, Authentication Header (AH) and Encapsulating Security Payload (ESP).

1. AH Protocol :

- The AH protocol provides a mechanism for authentication only. AH provides data integrity, data origin authentication, and an optional replay protection service. Data integrity is ensured by using a message digest that is generated by an algorithm such as HMAC-MD5 or HMAC-SHA. Data origin authentication is ensured by using a shared secret key to create the message digest. Data origin authentication is ensured by using a shared secret key to create the message digest. Replay protection is provided by using sequence number field with the AH header.

Data used in calculation of authentication data
(except those fields in IP header changing during transmission)

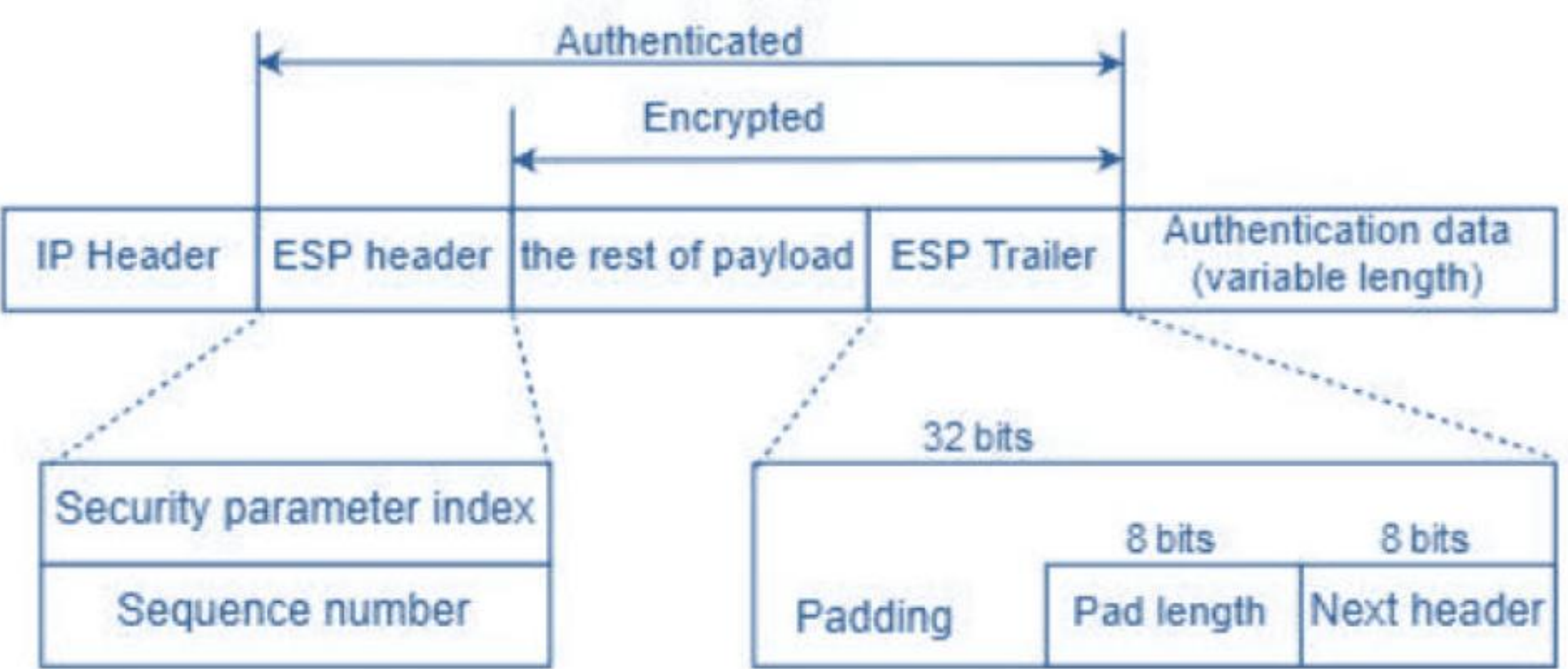


Authentication Header (AH) protocol

2. ESP Protocol :

- The ESP protocol provides data confidentiality (encryption) and authentication (data integrity, data origin authentication, and replay protection). ESP can be used with confidentiality only, authentication only, or both confidentiality and authentication. When ESP provides authentication functions, it uses the same algorithms as AH, but the coverage is different. AH-Style authentication authenticates the entire IP packet, including the outer IP header, while the ESP authentication mechanism authenticates only the IP datagram portion of the IP packet.

2. ESP Protocol :



ESP

Firewalls:

- A firewall is network security device, either h/w or s/w based, which monitors all incoming and outgoing traffic and based on a defined set of security rules it accepts, rejects or drops that specific traffic.
 - ❖ **Accept:** allow the traffic
 - ❖ **Reject:** block the traffic but reply with an “unreachable error”
 - ❖ **Drop:** block the traffic with no reply
- A firewall establishes a barrier between secured internal networks and outside untrusted network, such as the Internet.

Firewalls:

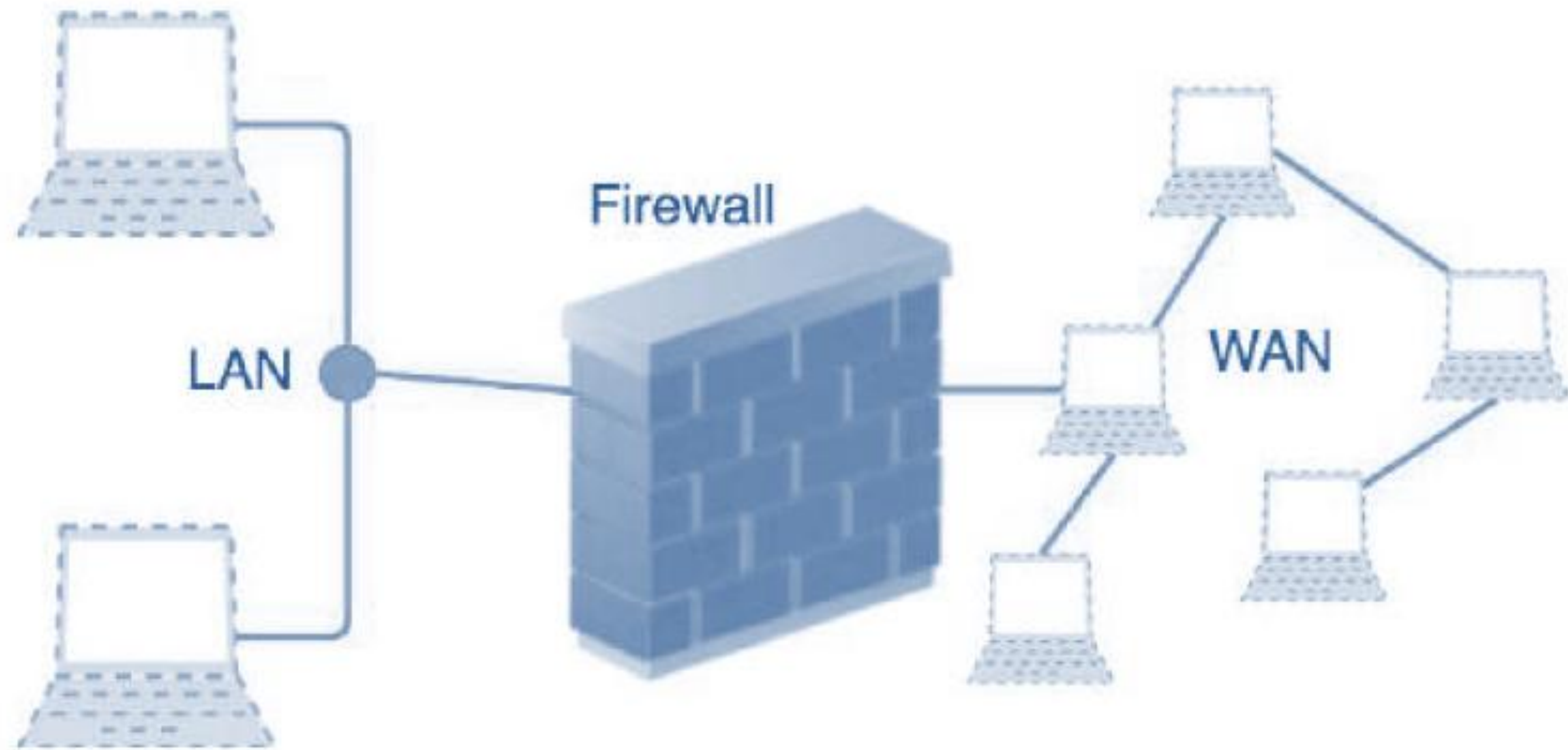


Fig: Firewall

Firewall Characteristics :

➤ Major characteristics related to firewall protection are described below:

1. Various protection levels
2. Wireless network (Wi-Fi) Protection
3. Internet and network access
4. Blockage against unauthorized access
5. Protection against malware
6. Provide access only to valid data packets
7. Provision of different configurations
8. Provision of numerous security policies
9.

Types of Firewalls :

- Firewalls come in various forms but there are two ways of understanding the types of firewalls.
 1. **Delivery:** This refers to where firewalls operate in the network context. Delivery options can include hardware, software, or cloud-based systems.
 2. **Technology:** This refers to how firewalls actually work. Firewall technology includes packet inspection, proxy or app-based firewalls, stateful firewalls, and next-generation systems.

Firewalls based on delivery method :

1. Hardware Based Firewall :

- Hardware firewalls are installed on special devices within your network rack. These devices contain firmware that creates a barrier between the external internet and on-premises systems. Inbound traffic must traverse this barrier, which provides a secure boundary to block malicious attacks.

2. Software Based Firewall :

- Software firewalls provide the same services as hardware-based firewalls but require no separate equipment. Instead, software firewalls run on network devices. For example, Microsoft Windows and macOS come with a native firewall to protect internet traffic.
- Users can install software-based firewalls on individual workstations. But they can also install them on servers to provide network-wide protection.

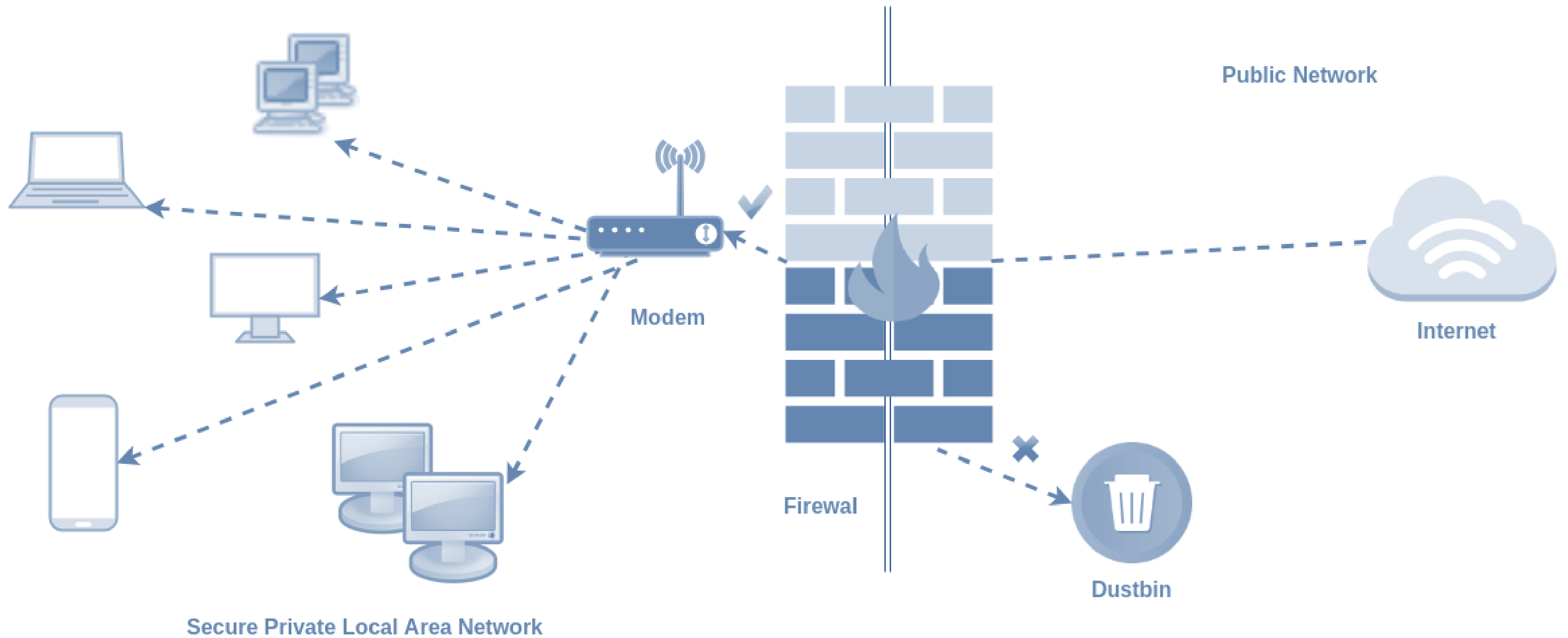
3. Cloud-Based Firewall :

- Software and hardware-based firewalls operate in traditional on-premises environments. They protect groups of devices against external threats. Cloud-based firewalls are different. Firewall-as-a-Service exists in the cloud and requires no extra hardware or software installation.
- Cloud firewalls primarily protect cloud assets. This includes Software-as-a-Service (SaaS) apps as well as Infrastructure-as-a-Service (IaaS) and Platform-as-a-Service (PaaS) implementations.
- At the same time, cloud firewalls extend protection to on-premises servers and remote work devices. Instead of using software or hardware, they employ cloud-based applications to monitor traffic and block attacks.

Firewalls Based on Method of Operation:

1. Packet Filtering Firewall :

- Packet-filtering firewalls assess data packets passing across network boundaries. Each packet must be compared to a set of pre-defined rules. If the packets meet these rules, the firewall allows traffic to pass. If not, packets are blocked and alerts may be issued.
- Information assessed by packet-filtering firewalls can include the destination port in use, the type of packet, and the destination IP address. This data provides a snapshot of where incoming packets come from, who is sending them, and whether they are safe.
- Packet filtering firewalls are generally installed at junctions within network environments. They may operate alongside switches and routers, processing traffic at high-volume locations. This enables packet-filtering firewalls to handle traffic for an entire network.

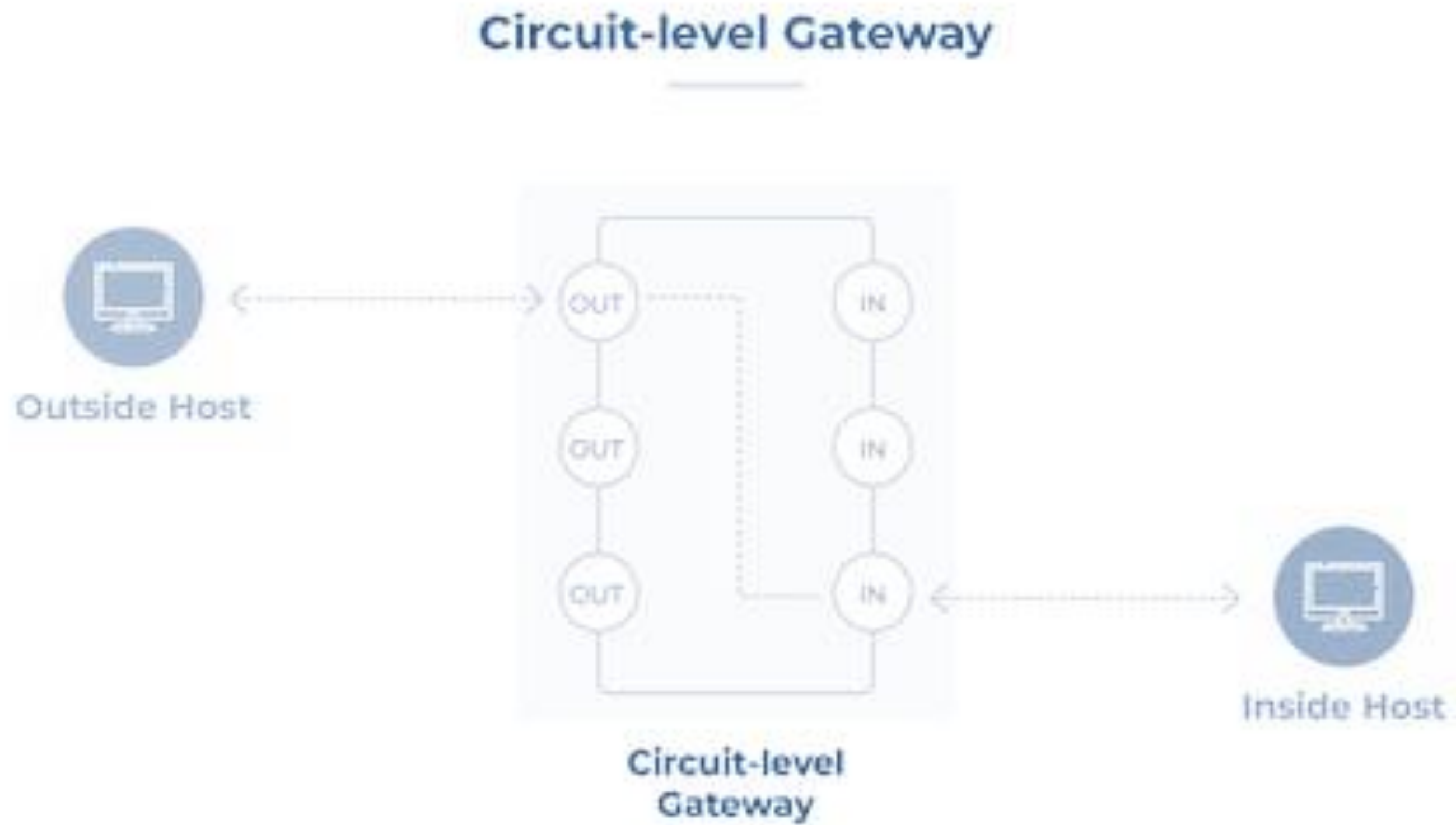


- ✓ = Specified Traffic Allowed
- ✗ = Restricted Unknown Traffic

2. Circuit-Level Gateways :

- Circuit level gateways operate at the session level. They assess traffic when local and remote hosts establish a connection. If this connection is deemed insecure, circuit level gateways will close and prevent communication between the two devices.
- Circuit level gateways use session information such as TCP handshaking and protocol messaging. By leveraging this information, the circuit level gateway can establish whether sessions are legitimate. There is no data packet inspection. Everything relates to the way connections are created.

2. Circuit-Level Gateways :



3. Stateful Inspection Firewalls :

- Stateful inspection firewalls add another level of sophistication to firewall protection. Standard firewalls are stateless. They make decisions based on inputs, with no further requests for information.
- Stateful firewalls take inputs and interrogate them. They leverage data from all network layers to establish context. Information about past connections is stored and mobilized to analyze future access requests.
- Contextual and historical information enables stateful firewalls to make informed decisions. In general, this results in more comprehensive threat protection in complex network environments.

4. Proxy Firewalls :

- Also known as web application or application layer firewalls, proxy firewalls route data packets through separate proxy servers.
- Proxy servers act as a network gateway between remote devices and web applications. Proxy firewalls screen incoming and outgoing traffic, blocking direct access to web servers without proper authentication.
- Proxy firewalls operate at the application layer. They filter by destination port, but also assess data packets using application data. For instance, the firewall may check HTTP request strings to determine whether connections are legitimate.
- Operating at the application layer allows security teams to apply controls over web usage. Admins can block access to content according to HTTP addresses. Granular controls also allow access to certain web resources while blocking insecure websites.

5. Next-generation Firewalls :

- Next-generation firewalls (NGFWs) build on other firewall types and add extra functionality. Generally, an NGFW will blend packet inspection with the contextual controls offered by stateful firewalls.
- NGFW services offer deep security controls to protect sensitive data. Deep Packet Inspection (DPI) analyzes the content of data packets, not just header information. This extra knowledge allows the firewall to verify the authenticity of HTTP transfers. The result is improved web security.
- Anti-malware and antivirus scanning extend protection further. NGFWs also commonly include intrusion prevention and detection systems (IDS/IPS). Intrusion prevention services detect possible threats and deliver alerts before attacks become critical.

Limitations of Firewall

1. Firewalls are not able to stop the users from accessing the data or information from malicious websites, making them vulnerable to internal threats or attacks.
2. It is not able to protect against the transfer of virus-infected files or software if security rules are misconfigured, against non-technical security risks (social engineering)
3. It does not prevent misuse of passwords and attackers with modems from dialing in to or out of the internal network.
4. Already infected systems are not secured by Firewalls.

!!!Thank you!!!